Measure Theoretic Probability
Lesson 5
Extension of Probability Measures: Proof Strategy and Uniqueness

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Outline

1. Recap: The Outer Measure
2. A Measurability Condition
3. Strategy of the Proof
4. Conclusion of the Extension
5. Uniqueness of the Extension
6. Summary and Outlook

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Setting and Definition of $\mathbb{P}^*$

Setting

Let Ω be a non-empty set, let $\mathcal{F}_0$ be a field on Ω, let $\mathcal{F} = \sigma(\mathcal{F}_0)$, and let $\mathbb{P}$ be a probability measure on the field $\mathcal{F}_0$.

Outer measure

For every subset $A \subseteq \Omega$, define

$$\mathbb{P}^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mathbb{P}(A_n) : \begin{array}{l} A_n \in \mathcal{F}_0 \text{ for all } n, \\ A \subseteq \bigcup_{n=1}^{\infty} A_n \end{array} \right\}.$$

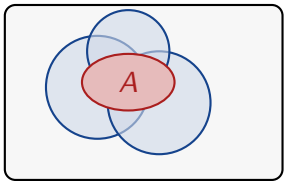
Goal

Show that $\mathbb{P}^*$, restricted to $\mathcal{F}$, is a probability measure extending $\mathbb{P}$; that is,

$$\mathbb{P}^*(A) = \mathbb{P}(A), \qquad A \in \mathcal{F}_0.$$

Remarks on the Covering Definition

- Covers may **overlap**, causing the covering sum to count some parts more than once.
- A cover may also include points outside A. Thus, each covering sum can overestimate the size of A.
- A cover may require countably many sets. Their union need not belong to $\mathcal{F}_0$, so we use the sum of probabilities rather than the probability of the union.
- Disjoint covers and covers consisting of a single set from $\mathcal{F}_0$ are special cases of the general covering definition.



Overlapping covering sets from $\mathcal{F}_0$.

Why "Outer" Measure?

Terminology

The outer measure $\mathbb{P}^*$ approaches a set A from the **outside**, using sets that contain A. An inner measure could instead be defined by using sets contained in A and taking the largest possible value.

Why not simply call it a measure?

A measure is defined on a measurable space $(\Omega, \mathcal{G})$, where $\mathcal{G}$ is a chosen σ-field.
At this stage, we have not yet established that $\mathbb{P}^*$ is a measure on a particular σ-field. Our goal is to establish this property on $\mathcal{F} = \sigma(\mathcal{F}_0)$.

What Is Already Known About $\mathbb{P}^*$

Established in Lesson 4

Property	Statement
Null set	$\mathbb{P}^*(\emptyset) = 0$
Monotonicity	$A \subseteq B \subseteq \Omega \Rightarrow \mathbb{P}^*(A) \leq \mathbb{P}^*(B)$
Countable subadditivity	$\mathbb{P}^*(\bigcup_n A_n) \leq \sum_n \mathbb{P}^*(A_n)$ for arbitrary subsets A_n
Agreement on $\mathcal{F}_0$	$\mathbb{P}^*(A) = \mathbb{P}(A)$ for $A \in \mathcal{F}_0$
Total mass	$\mathbb{P}^*(\Omega) = \mathbb{P}(\Omega) = 1$, with $0 \leq \mathbb{P}^*(A) \leq 1$

Important distinction

The inequality $\mathbb{P}^*(A) \leq \mathbb{P}(A)$ was established first. The reverse inequality, needed for equality on $\mathcal{F}_0$, is revisited in Lemma 5.

What Remains to Be Shown

Requirements for a probability measure on $\mathcal{F}$

Requirement
$\mathbb{P}^*(\emptyset) = 0$
$\mathbb{P}^*(\Omega) = 1$, values in $[0, 1]$
$\mathbb{P}^* = \mathbb{P}$ on $\mathcal{F}_0$
Countable additivity on $\mathcal{F}$

The obstacle

Countable **subadditivity** gives an inequality for arbitrary sets. Countable **additivity** gives equality for pairwise disjoint measurable sets. The latter has not yet been established.

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A Necessary Identity for Probability Measures

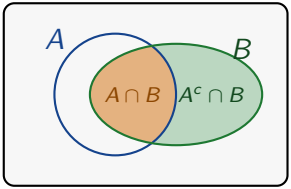
Observation

Let $\mathbb{Q}$ be a probability measure on a σ-field $\mathcal{G}$, and let $A, B \in \mathcal{G}$.
Split B into two disjoint sets:

$$B = (A \cap B) \cup (A^c \cap B).$$

By additivity,

$$\mathbb{Q}(B) = \mathbb{Q}(A \cap B) + \mathbb{Q}(A^c \cap B).$$



Question

Does the same identity hold with $\mathbb{P}^*$ in place of $\mathbb{Q}$? In general, not for every subset A.

The Class $\mathcal{M}$ of $\mathbb{P}^*$-Measurable Sets

Definition: Carathéodory measurability

Define

$$\mathcal{M} = \{A \subseteq \Omega : \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) = \mathbb{P}^*(B) \text{ for every } B \subseteq \Omega\}.$$

Equivalent definition

A set A belongs to $\mathcal{M}$ if and only if

$$\mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) \leq \mathbb{P}^*(B)$$

for every $B \subseteq \Omega$.

Why is this equivalent?

Since $B = (A \cap B) \cup (A^c \cap B)$, countable subadditivity gives

$$\mathbb{P}^*(B) \leq \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B).$$

Thus, the reverse inequality is automatic, and only the displayed inequality needs to be verified.

The Role of $\mathcal{M}$

Central idea

Rather than attempting to establish additivity for every subset of Ω, we restrict attention to sets for which the Carathéodory splitting identity holds. These sets form $\mathcal{M}$.

Trivial members of $\mathcal{M}$

Both $\emptyset$ and Ω belong to $\mathcal{M}$.
For $A = \emptyset$,

$$\mathbb{P}^*(\emptyset \cap B) + \mathbb{P}^*(\Omega \cap B) = 0 + \mathbb{P}^*(B).$$

The case $A = \Omega$ is symmetric. Moreover,

$$A \in \mathcal{M} \iff A^c \in \mathcal{M},$$

because the defining identity is symmetric in A and A^c.

Terminology

The class $\mathcal{M}$ is called the class of **Carathéodory-measurable sets**. It is defined using $\mathbb{P}^*$ rather than a σ-field chosen in advance.

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Proof Strategy: Six Lemmas

Why a roadmap?

The complete proof is long and spans several lessons. The lemmas provide a roadmap for the overall argument.

The lemmas

Lemma	Statement	The reason we do that
1	$\mathcal{M}$ is a field	Verify finite set operations
2	Pairwise disjoint $A_n \in \mathcal{M}$ satisfy $\mathbb{P}^*(\bigcup_n A_n) = \sum_n \mathbb{P}^*(A_n)$	Establish additivity
3	$\mathcal{M}$ is a σ-field	Extend closure to countable unions
4	$\mathcal{F}_0 \subseteq \mathcal{M}$	Place the original field inside $\mathcal{M}$
5	$\mathbb{P}^*(A) = \mathbb{P}(A)$ for $A \in \mathcal{F}_0$	Establish extension
6	$\mathbb{P}^*$ is a probability measure on $(\Omega, \mathcal{M})$	Complete the measure construction

Comments on the Lemmas

Lemma 3 needs more than Lemma 1

A field is closed under **finite** unions. A σ-field must also be closed under **countable** unions. The additivity established in Lemma 2 enables this step.

Lemma 2 versus subadditivity

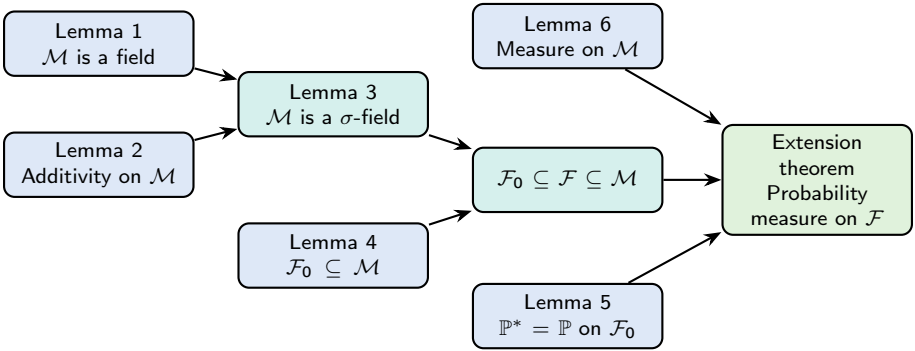
Subadditivity holds for arbitrary subsets and gives an inequality. Lemma 2 gives equality for pairwise disjoint sets in $\mathcal{M}$:

$$\mathbb{P}^*\left(\bigcup_n A_n\right) = \sum_n \mathbb{P}^*(A_n).$$

Lemma 5

This is the agreement $\mathbb{P}^* = \mathbb{P}$ on $\mathcal{F}_0$

Dependencies Among the Lemmas



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Step 1: $\mathcal{F}$ Is Sandwiched Between $\mathcal{F}_0$ and $\mathcal{M}$

Claim

Assuming Lemmas 3 and 4,

$$\mathcal{F}_0 \subseteq \mathcal{F} \subseteq \mathcal{M}.$$

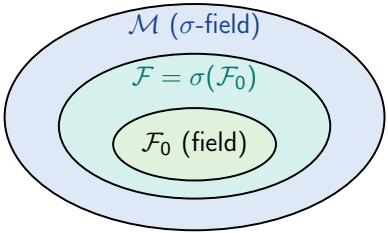
Proof

By definition, $\mathcal{F} = \sigma(\mathcal{F}_0)$, so $\mathcal{F}_0 \subseteq \mathcal{F}$.
Lemmas 3 and 4 establish that $\mathcal{M}$ is a σ-field containing $\mathcal{F}_0$. Since $\mathcal{F}$ is the smallest σ-field containing $\mathcal{F}_0$, it follows that

$$\mathcal{F} \subseteq \mathcal{M}.$$

Key principle

The generated σ-field is contained in every σ-field that contains the original field.



Step 2: The Extension Theorem

Theorem: Extension

Assume Lemmas 1–6. The restriction of $\mathbb{P}^*$ to $\mathcal{F} = \sigma(\mathcal{F}_0)$ is a probability measure on $(\Omega, \mathcal{F})$, and

$$\mathbb{P}^*(A) = \mathbb{P}(A), \qquad A \in \mathcal{F}_0.$$

Proof: Zero mass and additivity

Since $\mathcal{F} \subseteq \mathcal{M}$, the restriction of $\mathbb{P}^*$ to $\mathcal{F}$ is well-defined.

1. Since $\emptyset \in \mathcal{F}$,
$$\mathbb{P}^*(\emptyset) = 0.$$
2. If $A_n \in \mathcal{F}$ are pairwise disjoint, then $\bigcup_n A_n \in \mathcal{F}$. Since $A_n \in \mathcal{M}$, Lemma 2 gives
$$\mathbb{P}^*\left(\bigcup_n A_n\right) = \sum_n \mathbb{P}^*(A_n).$$

Step 2: The Extension Theorem (continued)

Proof: Total mass and agreement

Since $\Omega \in \mathcal{F}_0$, Lemma 5 gives

$$\mathbb{P}^*(\Omega) = \mathbb{P}(\Omega) = 1.$$

Agreement with $\mathbb{P}$ on $\mathcal{F}_0$ follows from Lemma 5. Together with the preceding slide, this establishes all required properties.

Theorem

This is the **Carathéodory extension theorem**.

Why Restriction to $\mathcal{F}$ Preserves the Measure Property

General principle

If μ is a measure on a σ-field $\mathcal{M}$ and $\mathcal{F} \subseteq \mathcal{M}$ is a smaller σ-field, then $\mu|_{\mathcal{F}}$ is a measure on $\mathcal{F}$.

Verification

- The empty set belongs to $\mathcal{F}$, and $\mu|_{\mathcal{F}}(\emptyset) = 0$.
- A countable union of pairwise disjoint sets in $\mathcal{F}$ remains in $\mathcal{F}$. Countable additivity therefore applies.
- The value of the restricted measure on any set in $\mathcal{F}$ is inherited from μ.

For probability measures

Because $\Omega \in \mathcal{F}$ and $\mu(\Omega) = 1$, the restriction also has total mass one.

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Uniqueness Theorem

Theorem

Let $\mathcal{F}_0$ be a field, let $\mathcal{F} = \sigma(\mathcal{F}_0)$, and let $\mathbb{P}^*$ be the extension constructed above.
If $\mathbb{Q}$ is any probability measure on $\mathcal{F}$ satisfying

$$\mathbb{Q}(A) = \mathbb{P}(A), \qquad A \in \mathcal{F}_0,$$

then

$$\boxed{\mathbb{Q}(A) = \mathbb{P}^*(A) \quad \text{for every } A \in \mathcal{F}.}$$

Interpretation

The extension is **unique**: the values of a probability measure on $\mathcal{F}_0$ determine its values on the entire generated σ-field $\mathcal{F}$.

Proof plan

1. Show that $\mathbb{Q}(A) \leq \mathbb{P}^*(A)$.
2. Apply the same inequality to A^c to obtain the reverse inequality.

Proof of Uniqueness: Step 1

Claim

For every $A \in \mathcal{F}$,

$$\mathbb{Q}(A) \leq \mathbb{P}^*(A).$$

Proof

Let $A_1, A_2, \ldots \in \mathcal{F}_0$ be any cover of A, so that

$$A \subseteq \bigcup_{n=1}^{\infty} A_n.$$

Because $\mathbb{Q} = \mathbb{P}$ on $\mathcal{F}_0$, countable subadditivity and monotonicity of $\mathbb{Q}$ imply

$$\begin{aligned} \sum_{n=1}^{\infty} \mathbb{P}(A_n) &= \sum_{n=1}^{\infty} \mathbb{Q}(A_n) \\ &\geq \mathbb{Q}\left(\bigcup_{n=1}^{\infty} A_n \right) \\ &\geq \mathbb{Q}(A). \end{aligned}$$

Proof of Uniqueness: Step 1 (continued)

Taking the infimum

Every covering sum is at least $\mathbb{Q}(A)$. Taking the infimum over all covers of A gives

$$\mathbb{Q}(A) \leq \inf \left\{ \sum_{n=1}^{\infty} \mathbb{P}(A_n) : A \subseteq \bigcup_{n=1}^{\infty} A_n,\ A_n \in \mathcal{F}_0 \right\}.$$

By the definition of outer measure,

$$\boxed{\mathbb{Q}(A) \leq \mathbb{P}^*(A).}$$

Proof of Uniqueness: Step 2

Proof of the reverse inequality

Let $A \in \mathcal{F}$. Since $\mathcal{F}$ is a σ-field, $A^c \in \mathcal{F}$.
Applying Step 1 to A^c gives

$$\mathbb{Q}(A^c) \leq \mathbb{P}^*(A^c).$$

Both $\mathbb{Q}$ and $\mathbb{P}^*$ are probability measures on $\mathcal{F}$. Using the complement rule,

$$1 - \mathbb{Q}(A) \leq 1 - \mathbb{P}^*(A).$$

Rearranging,

$$\boxed{\mathbb{P}^*(A) \leq \mathbb{Q}(A).}$$

Proof of Uniqueness: Final Step

Combine the inequalities

Step 1 gives $\mathbb{Q}(A) \leq \mathbb{P}^*(A)$, while Step 2 gives $\mathbb{P}^*(A) \leq \mathbb{Q}(A)$. Therefore,

$$\boxed{\mathbb{Q}(A) = \mathbb{P}^*(A), \qquad A \in \mathcal{F}.}$$

Important

The argument uses the fact that $\mathbb{P}^*$ is a probability measure on $\mathcal{F}$, established by the extension theorem.

A Stronger Uniqueness Statement

Proposition

Let $\mathcal{F}_0$ be a field and let $\mathcal{F} = \sigma(\mathcal{F}_0)$.
If $\mathbb{P}_1$ and $\mathbb{P}_2$ are probability measures on $\mathcal{F}$ and

$$\mathbb{P}_1(A) = \mathbb{P}_2(A), \qquad A \in \mathcal{F}_0,$$

then

$$\mathbb{P}_1(A) = \mathbb{P}_2(A), \qquad A \in \mathcal{F}.$$

Comments

- This result does not require the outer measure construction. It applies to any two probability measures on $\mathcal{F}$.
- A stronger version holds when $\mathcal{F}_0$ is a π-**system**, a concept to be introduced later.

Book

A standard reference is Billingsley, *Probability and Measure*

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Summary

Main ideas

Object / result	Content
Outer measure $\mathbb{P}^*$	Infimum of covering sums over sets from $\mathcal{F}_0$; defined on all subsets of Ω
Class $\mathcal{M}$	Sets A satisfying the Carathéodory splitting identity for every subset B
Lemmas 1–6	Establish that $\mathcal{M}$ is a σ-field containing $\mathcal{F}_0$, on which $\mathbb{P}^*$ is a probability measure
Extension theorem	$\mathcal{F}_0 \subseteq \mathcal{F} \subseteq \mathcal{M}$, so $\mathbb{P}^*\|_{\mathcal{F}}$ is a probability measure extending $\mathbb{P}$
Uniqueness	Any probability measure on $\mathcal{F}$ agreeing with $\mathbb{P}$ on $\mathcal{F}_0$ equals $\mathbb{P}^*$

Central conclusion

A probability measure on a field can be extended to the generated σ-field, and the extension is unique.

Exercises

1. Show that $\emptyset, \Omega \in \mathcal{M}$ and that $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$.
2. Prove that, for all $A, B \subseteq \Omega$,

$$\mathbb{P}^*(B) \leq \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B).$$

3. Explain why

$$1 - \mathbb{Q}(A) \leq 1 - \mathbb{P}^*(A) \quad \Longrightarrow \quad \mathbb{P}^*(A) \leq \mathbb{Q}(A).$$

4. Let μ be a measure on a σ-field $\mathcal{M}$ and let $\mathcal{F} \subseteq \mathcal{M}$ be a sub-σ-field. Verify that $\mu|_{\mathcal{F}}$ is a measure on $\mathcal{F}$.

Outlook

Next lessons

- Prove Lemmas 1–3, showing that $\mathcal{M}$ is a σ-field on which $\mathbb{P}^*$ is additive.
- Prove Lemmas 4–6, establishing that $\mathcal{F}_0 \subseteq \mathcal{M}$ and that $\mathbb{P}^*$ is a probability measure extending $\mathbb{P}$.

After the extension theorem

The next topic concerns limits of sequences of sets, including the **limit inferior** and **limit superior** of a sequence of sets.